ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 1/2	Offspring Tt and tt in 1 : 1.
2	**(b)** fertilisation	The sperm type decides.
3	**(a)** both alleles are recessive	Only tt shows dwarf.
4	**(a)** both Tt	3 : 1 ratio arises from Tt x Tt.
5	**(a)** 1/2	Each birth is independent.
6	**(c)** minor DNA copying errors	Copying errors create small differences.
7	**(c)** 40	1/4 of 160.
8	**(d)** Sperm decides the sex	The father's gamete is X or Y.
9	**(b)** an acquired trait due to environment	Environment affects the phenotype.
10	**(c)** TT	A homozygous genotype forms a single type of gamete.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Dwarf offspring need a t allele from the tall parent (1); Tall parent is Tt (1).	1+1
12	Wrinkled (rr) are 1/4 (1); 100 seeds (1).	1+1
13	Round = 9 + 3 = 12 out of 16 (1); 3/4 (1).	1+1
14	F1 plants are Tt carrying the t allele (1); In F2 two t alleles may combine giving tt (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	F1 Tt; F2 ratio 3 : 1 (1); Tall = 225, dwarf = 75 (1); Genotypes TT 75, Tt 150, tt 75 (1).	1+1+1
16	Round yellow 9/16 = 450 (1); Round green 3/16 = 150, wrinkled yellow = 150 (1); Wrinkled green 1/16 = 50 (1).	1+1+1
17	Blending would give intermediate seeds (1); Mendel found all F1 were round yellow (1); F2 showed original traits again, rejecting blending (1).	1+1+1
	OR: Gametes R, r and R, r (1); Offspring RR, Rr, Rr, rr (1); Genotypic 1 : 2 : 1 and phenotypic 3 round : 1 wrinkled (1).	1+1+1
18	Environment can change the phenotype (1); Plants: a genetically tall plant grows short in poor soil (1); Humans: tan or body weight depends on conditions (1).	1+1+1
	OR: Expected ratio 3 : 1 (1); Tall about 15, dwarf about 5 (1); The ratio holds well for large numbers of offspring (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Gametes RY, Ry, rY, ry (1); 16 combinations (1); 9 round yellow (1); 3 round green and 3 wrinkled yellow (1); 1 wrinkled green (1).	1+1+1+1+1
	OR: Genotypes TT, Tt and tt (1); Phenotypes tall and dwarf (1); T is dominant (1); t is recessive (1); T and t are alleles (1).	1+1+1+1+1
20	Parents TTRR x ttrr (1); F1 is TtRr, tall round (1); Tall round 90 (1); Tall wrinkled 30 and dwarf round 30 (1); Dwarf wrinkled 10 (1).	1+1+1+1+1
	OR: Man is ff (1); Woman is Ff (her mother gave f) (1); Offspring Ff and ff (1); Half free and half attached earlobes (1); Ratio 1 : 1 (1).	1+1+1+1+1

Mistakes to avoid:

- Giving 1 : 2 : 1 as the F2 *phenotypic* ratio of a monohybrid cross; it is 3 : 1 phenotypic and 1 : 2 : 1 genotypic.
- Saying that the dominant trait is 'stronger' or 'more common'; it simply masks the recessive allele in the heterozygote.
- Writing the genotype of F1 as TT or tt; the cross of pure TT and pure tt gives only Tt.
- Saying that the mother decides the sex of the child; all eggs carry X, and the father's sperm (X or Y) determines it.
- Forgetting that a recessive trait is expressed only in the homozygous recessive (tt), so it disappears in F1 but reappears in F2.

- Writing the number of chromosomes of the sex cells as 46; gametes contain 23 chromosomes (22 autosomes + 1 sex chromosome).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 23	22 + 1 = 23.
2	**(a)** short	The genotype is unchanged.
3	**(d)** 4	RY, Ry, rY, ry.
4	**(b)** 22	22 autosomes plus one X.
5	**(a)** 1 : 2 : 1	TT : Tt : tt.
6	**(d)** 27	3/16 of 144 = 27.
7	**(c)** 44	22 pairs.
8	**(b)** 1/2	Tt in the middle, 2 out of 4.
9	**(c)** 1/16	Only rryy.
10	**(d)** Sperm	Gametes are haploid.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Parental combinations (round yellow, wrinkled green) appear in F2 (1); New combinations (round green, wrinkled yellow) also appear (1).	1+1
12	Both parents are heterozygous Ff (1); The child gets f from each parent: ff (1).	1+1
13	A woman has only X chromosomes (XX) (1); Her eggs carry only X (1).	1+1
14	Sex depends on the sperm (1); Her eggs carry only X (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	About half the offspring are green (1); Yellow parent must give y gametes (1); Yellow parent is Yy (1).	1+1+1
16	X and Y sperms are produced in equal numbers (1); Each has the same chance of fertilising the egg (1); Hence half the children are boys and half girls (1).	1+1+1
17	A gene codes for an enzyme (1); The enzyme helps make the plant growth hormone (1); Efficient enzyme gives tall, less efficient gives dwarf (1).	1+1+1
	OR: Traits coded in germ-cell DNA are inherited, e.g. eye colour (1); Traits gained during life affect only body cells (1); These are not inherited, e.g. a scar (1).	1+1+1
18	Monohybrid: one character; dihybrid: two characters (1); F2 ratio: 3 : 1 and 9 : 3 : 3 : 1 (1); F1 is all dominant in both (1).	1+1+1
	OR: The child gets 23 chromosomes from the mother (1); 23 from the father (1); So traits are a mix of both (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Unknown tall may be TT or Tt (1); Cross with tt (1); If TT, all offspring are tall (1); If Tt, half are dwarf (1); So the test cross reveals the genotype (1).	1+1+1+1+1
	OR: DNA copying errors cause variations (1); Daughter cells inherit them (1); Asexual reproduction gives small variations (1); Sexual reproduction mixes two sets of genes (1); So it gives more variation (1).	1+1+1+1+1
20	Mother XX produces only X eggs (1); Father XY produces X and Y sperms equally (1); X sperm gives a girl and Y sperm gives a boy (1); Chances are equal for boy and girl (1); The mother cannot influence it as the father's sperm decides (1).	1+1+1+1+1
	OR: Body cell 46 chromosomes in 23 pairs (1); Gametes have 23 (1); Zygote regains 46 (1); 22 pairs are autosomes, one pair sex chromosomes (1); Mother gives X; father gives X or Y (1).	1+1+1+1+1

Mistakes to avoid:

- Giving 1 : 2 : 1 as the F2 *phenotypic* ratio of a monohybrid cross; it is 3 : 1 phenotypic and 1 : 2 : 1 genotypic.

- Saying that the dominant trait is 'stronger' or 'more common'; it simply masks the recessive allele in the heterozygote.
- Writing the genotype of F1 as TT or tt; the cross of pure TT and pure tt gives only Tt.
- Saying that the mother decides the sex of the child; all eggs carry X, and the father's sperm (X or Y) determines it.
- Forgetting that a recessive trait is expressed only in the homozygous recessive (tt), so it disappears in F1 but reappears in F2.
- Writing the number of chromosomes of the sex cells as 46; gametes contain 23 chromosomes (22 autosomes + 1 sex chromosome).

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** TT x tt	F1 will all be Tt.
2	**(c)** 1 : 1 : 1 : 1	Test cross for two traits.
3	**(a)** one X to each child	Her eggs carry one X.
4	**(c)** Tt x tt	Tt and tt in 1 : 1.
5	**(a)** round green	Round green and wrinkled yellow are new combinations.
6	**(a)** all his daughters	He gives X only to daughters.
7	**(a)** both heterozygous free earlobes	Both must give an f gamete.
8	**(a)** traits of parents blend in offspring	Recessive traits reappear intact in F2.
9	**(b)** all his sons	Y goes only to sons.
10	**(d)** Seed shape in peas	Mendel's traits are gene controlled.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is recessive and masked in heterozygotes (1); It appears when homozygous recessive (1).	1+1
12	Homozygous TT and tt = 1/2 (1); 200 plants (1).	1+1
13	Cross with dwarf (tt) (1); Homozygous gives all tall, heterozygous gives half dwarf (1).	1+1
14	The zygote has XY (1); The child is a boy (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Egg: X; sperm: X or Y (1); XX (girl) and XY (boy) (1); Ratio 1 : 1 (1).	1+1+1
16	Gametes R and r from each parent (1); Combination rr arises with 1/4 probability (1); Probability = 25% (1).	1+1+1
17	Every egg and sperm has a different set of genes (1); The fusion is random (1); Hence each child is unique (1).	1+1+1
	OR: F1 hides the recessive factor (1); F2 shows it again in 1/4 (1); Pairs must separate in gametes and rejoin randomly (1).	1+1+1
18	DNA makes a copy during cell division (1); The copy is passed to daughter cells and gametes (1); Errors create new variations (1).	1+1+1
	OR: Person is heterozygous (1); The dominant allele masks the recessive (1); The recessive allele may be passed on (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Inherited: controlled by genes, passed on (1); Acquired: gained in life, not passed on (1); Example: a mouse whose tail is cut, its offspring have normal tails (1); Germ-cell DNA remains unchanged (1); A gene for an inherited trait is passed through the gamete (1).	1+1+1+1+1
	OR: Gametes: RY, Ry, rY, ry (1); 16 combinations (1); 4 phenotypes: round yellow, round green, wrinkled yellow, wrinkled green (1); 9 genotypes (1); Ratio 9 : 3 : 3 : 1 (1).	1+1+1+1+1
20	F1 plants were all tall (1); No intermediate height was seen (1); F2 had tall and dwarf in 3 : 1 (1); Dwarf trait reappeared unchanged (1); So factors are separate and not blended (1).	1+1+1+1+1
	OR: Cross 1: RRYY x rryy (1); F1 RrYy all round yellow (1); Cross 2: RRYy x rryy (1); Offspring RrYy and Rryy (1); Round yellow : round green = 1 : 1 (1).	1+1+1+1+1

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- Forgetting that a recessive trait is expressed only in the homozygous recessive (tt), so it disappears in F1 but reappears in F2.
- Writing the number of chromosomes of the sex cells as 46; gametes contain 23 chromosomes (22 autosomes + 1 sex chromosome).

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